MAYANK BANSAL

Machine Learning Engineer

Hisar, Haryana, India | +91 8708574164 | bansalmayank1414@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Third-year Computer Science student at LPU specializing in applied machine learning and real-world system development. Built and deployed multiple end-to-end ML systems across NLP, computer vision, and vector search, including a RAG-based question answering system and a visual search engine. Experienced in designing scalable pipelines, working with embeddings, and integrating ML models into production-ready APIs. Published researcher with a 9.33 CGPA and 580+ DSA problems solved.

EXPERIENCE

Machine Learning Intern | Qriocity Ventures

June 2026

- Developed deep learning models for medical image analysis using PyTorch, TIMM, and Computer Vision techniques.
- Built and trained a U-Net (ResNet34) for pneumothorax segmentation (Dice Score: 0.78) and an EfficientNet-B4 system for multi-class chest disease classification (weighted F1: 0.91).

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, CNNs, Neural Networks, Feature Engineering, Model Evaluation, EDA

Vector Search & RAG: FAISS, Sentence Transformers, Retrieval Augmented Generation

Backend & Deployment: Flask, FastAPI, REST APIs, Docker

Data & Languages: Python, NumPy, Pandas, Matplotlib, Seaborn, SQL

PROJECTS

AI Complaint Triage & Routing System - **Python, Flask, Scikit-learn, TF-IDF, SQLite** | **GitHub**

Jan 2026 – Feb 2026

- Automated complaint routing across 8 departments with urgency prediction using a hybrid credibility scoring pipeline combining rule-based logic and ML confidence scores
- Designed and published an 800-sample multi-class complaint dataset on Kaggle; built a modular Flask REST API with explainable decision flow and a foundation for continuous retraining

Visual Search Engine - **PyTorch, FAISS, FastAPI, JavaScript** | **GitHub**

Feb 2026 – Mar 2026

- Built an image-based retrieval system using a pretrained ResNet50 feature extractor generating 2048-dim embeddings and FAISS nearest-neighbour search over a fashion dataset, delivering real-time visual similarity results
- Engineered a FastAPI backend integrating image preprocessing, embedding generation, and vector search with a responsive upload-and-preview frontend — full end-to-end deployment

Multi-Hop Technical Debug Assistant - FastAPI, FAISS, Sentence Transformers, Groq (Llama 3.3) | **GitHub**

Jun 2026 – Jul 2026

- Developed a production-style Multi-Hop Retrieval-Augmented Generation (RAG) technical debugging assistant by implementing a custom retrieval pipeline with metadata-aware chunking, semantic embeddings, persistent FAISS indexing, multi-document retrieval, context assembly, and prompt engineering.
- Built an interactive FastAPI application supporting live knowledge base uploads, grounded LLM responses using Llama 3.3 via Groq, source attribution, semantic retrieval scores, and real-time performance metrics.

ACHIEVEMENTS & CERTIFICATIONS

- **Published a technical book —** "Machine Learning Projects: A Practical Guide for B.Tech CSE Students," on Amazon Kindle and Paperback, covering 6 end-to-end ML projects across NLP, CNNs, Transfer Learning, and Audio ML (https://www.amazon.com/dp/B0GX34T5HF)
- **Published research paper —** "Mood2Mail: A Lightweight Real-Time Email Tone Analyzer Using Classical ML" on Zenodo
- **Competitive programming —** 580+ problems on GeeksForGeeks (101-day streak, rank 159 / 39,000+ participants) + 100+ on LeetCode
- **Certifications —** Complete Data Science & ML, GeeksForGeeks (Jun–Nov 2025) - AI Tools Skill Up, GeeksForGeeks (Jun–Aug 2025)

EDUCATION

Lovely Professional University — B.Tech Computer Science Engineering Aug 2024 – May 2028

CGPA: 9.33/10 - Phagwara, Punjab, India